Pranay Katari

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Education

The University of Texas at Austin — Austin, TX
Bachelor of Science in Computer Science — *May 2028*

- Coursework: Data Structures, Discrete Mathematics, Computer Architecture, Linear Algebra, Statistics, Operating Systems, Algorithms, Machine Learning

Experience

Software Engineer Intern — June 2026 – August 2026
Invesco Ltd. — *Houston, TX*

- Automated **AWS EC2 gateway provisioning** with **Terraform** and **Bitbucket Pipelines** to support governed dashboard access for an EMEA audience of **5,000+ users**.
- Developed a **Python AWS Lambda** integration using **Secrets Manager**, gateway APIs, and **Snowflake**, supporting automated ServiceNow request processing through dynamic stored procedures and reducing **request fulfillment time by 2 days**.
- Built **Snowpark (Python)** scripts that aggregated **Snowflake Cortex** usage data for the data analytics team, supporting management's visibility into how employees were using AI.

Software Developer Fellow — September 2025 – Present
Longhorn Developers — *Austin, TX*

- Developed **React** and **TypeScript** features for **Degree Audit Plus**, a UT degree-planning **Chrome extension (Manifest V3)** built with **WXT**, **Vite**, and **Tailwind CSS**, serving **50,000+** users.
- Expanded the degree parser from 3 narrow patterns to 6 canonical formats plus a general fallback, adding B.A. support alongside B.S. so the tool works across more majors.
- Partnered with product managers to turn **Figma** designs into extension UI via the **Figma MCP** server, and prototyped analytics features incorporated into the final design.

Software Engineer Intern — May 2025 – September 2025
CodeSprout — *Austin, TX*

- Integrated the **Piston API** with **Dockerized sandboxes** to execute user-submitted code in isolated environments.
- Designed **PostgreSQL** schemas and indexes for frequently executed application queries.
- Developed AI-assisted student features that received **80% positive teacher feedback** in beta demonstrations.

Projects

Aquarius Baskets | Live Demo

- Implemented a **FastAPI/Pandas portfolio backtester** for **2–25 assets**, accounting for fractional shares, rebalancing, transaction costs, turnover, drawdown, and benchmark returns.
- Implemented **3-fold walk-forward validation**, benchmarking optimized portfolios against **SPY** and equal-weight baselines on out-of-sample returns, win rate, and drawdown.
- Built a **LangGraph** portfolio copilot with **pgvector** retrieval and persisted chat history to generate allocation-constrained basket proposals.
- Deployed a **Next.js/FastAPI** app on **Google Cloud Run** with Supabase OAuth, **Redis** caching, and persistent portfolios.

Real-Time BTC Event Processing Engine | Link

- Built a stateful **C++20 streaming gateway** processing Bitcoin market updates at 1-second intervals across **9 live assets**, with a measured **6.6 MB RSS** memory footprint.
- Normalized and reconciled market events from **4 live WebSocket/REST sources**, maintaining rolling volatility, velocity, acceleration, spread, and regime signals.
- Exposed streaming signals through asynchronous **FastAPI** endpoints to a live **Next.js** dashboard, with the Python data service using **44.7 MB RSS**.

Palladium Market Intelligence Agent | Link

- Trained a multi-task PyTorch LSTM on 60-day OHLCV sequences to predict price direction, 5-day realized volatility, and candle anomalies using chronological validation and early stopping.
- Built parallel **LangGraph** agents (OpenAI API, Tavily) analyzing supply, demand, and macro news, with an LLM judge synthesizing the findings into market research.

Dynamic Memory Allocator | *Course project, C*

- Implemented a **C** memory allocator over `sbrk` with **segregated free lists**, 16-byte alignment, block splitting, and coalescing, with constant-time free-list insertion and removal.

Technical Skills

Languages: Python, C++, TypeScript, SQL, Java, C
Frameworks / Data: FastAPI, React, Next.js, PyTorch, PostgreSQL, Redis, Snowflake
Cloud / Tools: AWS, GCP, Docker, Terraform, Git, Linux, LangGraph